Structure of the model

Six blocs, simulated year by year, 2026 to 2075, as a Monte Carlo over all uncertain parameters. Labels in brackets refer to the assumptions in Section 4 of the paper.

Scenario tree around the bloc simulation

Earlier stages (US and China, regimes still staffed by humans): automation race → collapse of broad economic participation → broad repression.
Competing end points: AI takeover not directed by rulers, non-AI global catastrophe (a run that hits one first is not counted).
Counterfactual: if the US becomes broadly repressive first, all six blocs continue from a world where the US is autocratic at closure.

Inside each bloc (US, China, Europe plus, Russia and MENA, South Asia, Global South). Each bloc is modeled as one state with one ruling coalition.

1 AI and robots

AI capability follows the measured task-horizon trend and can speed up (A1). A bloc's own chips and energy add capability; other blocs lag and catch up. Robots build robots, limited by energy, chips, mining and world inputs (A2, B6).

2 Automation and closure

Nine economic segments automate. A dedicated automated core loop can be built beside the old economy.
Human-dependence index: share of 2026 labor still needed.
CLOSURE when it falls below 20%.

3 Force

Security automates first (A3). Once fielded at scale, refusal by soldiers and police stops mattering (A4).
Military power = spending x output x automated share. Power inside the ruling group = control of that force (A6).

4 Democracy

V-Dem liberal democracy score. Falls as labor and revolt leverage fall. Below 0.30 = breakdown. Sets the decision rule in box 7.

5 The public's leverage

Labor: people can stop working.
Revolt: people can fight.
Enforcers: soldiers can refuse.
All three fall with closure (box 2) and automated force (box 3).

6 Grievance and unrest

Displacement and provision cuts (no safety net by default, A5) → grievance → hostility → yearly insurgency hazard → repression. Provision only partly buys peace.

threat

7 The ruling coalition decides

Every year once the bloc reaches closure. Up to 21 members (autocracy) or a 51-person elite (democracy), each with:
- moral cost of ordering killing (lower at the top in autocracies, B1)
- concern for the public, discount rate
- share of automated force (box 3)

Each member scores six options:

\+ public's welfare x concern
− cost of provision (falls with growth)
− threat of revolt and sabotage
− outside pressure if enforceable (B3)
\+ value of freed land and resources
− own moral cost of killing or neglect

Decision rule depends on regime:

democracy: 3 to 5 veto players
oligarchy: force-weighted supermajority
personalist: leader decides, inner circle may veto

Inside the group:

purges and counter-coups (B2). Brakes: fear of being next, leader mortality, protective doctrine. Objectors risk being purged.

8 Options (mild → harsh)

1 serve the public
2 live off rents
3 keep the status quo
4 warehouse (minimal provision)
5 neglect
6 depopulate

fear of survivors (A11, B4)

9 How deaths happen

Ordered depopulation: automated force, economic strangulation, abstract means of mass killing (A8)
Chosen, with no order to kill: lethal neglect
War, up to nuclear exchange (A9); attacks on weaker blocs (B5)
Total loss only: despair, crackdowns

What stops a campaign

a successful revolt, a repelled attack, or nuclear retaliation